III. Predict the output of the following:

1. String str = "Computer Applications" + 1 + 0;

   System.out.println("Understanding" + str);

2. String name = "Mr. Gupta";

   int age = 45;

   System.out.println(name + "is" + age + "years old");

3. String n1 = "46", n2 = "64";

   int total = Integer.parseInt(n1) + Integer.parseInt(n2);

   System.out.println("The sum of " + "46" + "and" + " 64" + "is" + total);

4. boolean p;

   p = ("BLUEJ".length() > "bluej".length()) ? true: false;

5. String str = "Information Technology";

   int p;

   p = str.indexOf('n');

   System.out.println(p);

6. char ch;

   int p=10;

   ch ="ICSE".charAt(0);

   System.out.println(ch + ":" + p);

7. String str1 = "Information Technology";

   String str2 = "information technology";

   boolean p = str1.equalsIgnoreCase(str2);

   System.out.println("The result is" + p);

8. What is the output of the following?

   char c = 'A';

   Short m = 26;

   int n = c +m;

   System.out.println(n) [ICSE 2005]

9. What do the following functions return for:-

   String x = "Hello";

   String y = "World";

   (a) System.out.println (x + y);

   (b) System.out.println (x.length ());

   (c) System.out.println (x.charAt(3));

   (d) System.out.println(x.equals(y)); [ICSE 2005]

10. What will be the output for the following program segment?

    String s = new String("abc");

    System.out.println(s.toUpperCase()); [ICSE 2006]

11. What is the output of the following:

    (i) System.out.println("four :"+ 4 + 2);

    System.out.println("four :"+ (2 + 2)); [ICSE 2007]

12. String S1 = "Hi";

    String S2 = "Hi";

    String S3 = "there";

    String S4 = "HI";

    System.out.println(S1 + "equals"+ S2 + " ⟶ " + S1.equals(S2));

    System.out.println(S1 + "equals"+ S3 + " ⟶ " + S1.equals(S3));

    System.out.println(S1 + "equals"+ S4 + " ⟶ " + S1.equals(S4));

    System.out.println(S1 + "equalsIgnoreCase"+ S4 + " ⟶ " +

    S1.equalsIgnoreCase(S4)); [ICSE 2007]

13. Predict the output:

    (i) char ch = "computer".charAt(3);

    System.out.println(ch);

    (ii) String str = "Rajesh";

    System.out.println(str.replace('j', 'k');

14. If String x = "Computer";
String y = "Applications";
What do the following functions return for:
(i) System.out.println(x.substring(1,5));
(ii) System.out.println(x.indexOf(x.charAt(4)));
(iii) System.out.println(y + x.substring(5));
(iv) System.out.println(x.equals(y));

[ICSE 2008]

IV. Differentiate the following:

1. equals() and = =
2. concatenation and truncation [ICSE 2008]
3. compareTo() and equals()
4. toLowerCase() and toUpperCase() [ICSE 2006]
5. charAt() and substring() [ICSE 2005]
6. Character.isLowerCase() and Character.tolowerCase()

V. Describe the purpose of the following functions with reference to Java programming. Write down the syntax of each of the following with an example.

1. toUpperCase()
2. trim()
3. toLowerCase()
4. length()
5. replace()
6. compareTo() [ICSE 2010]
7. reverse()
8. indexOf() [ICSE 2010]
9. startWith()
10. equalsIgnoreCase()

VI. Java Programming:

1. Write a program to accept two different characters and display the sum and difference of their ASCII values.

   Sample Input: A
   d

   Sample Output: The sum of ASCII values = 165
   The difference of ASCII value = 35

2. Write a program to input an alphabet in upper case or in lower case. Display the next alphabet accordingly. (i.e.: 'a' follows 'b', ... 'z' follows 'a').

3. Write a program to accept a character. If it is a letter, then display the case (i.e.: lower or upper case), otherwise check whether it is a digit or a special character.

   Sample Input: p

   Sample Output: p is a letter.
   p is in lower case.

4. Write a program in Java to accept a word/a String and display the new string after removing all the vowels present in it.

   Sample Input: COMPUTER APPLICATIONS

   Sample Output: CMPTR PPLCTNS

5. Write a program in Java to accept a word/a string. Count all the alphabets excluding vowels present in the word/string and display the result.

   Sample Input: Happy New Year

   Sample Output: No. of alphabets excluding vowels = 8

6. Write a program in Java to accept a name(Containing three words) and Display only the initials (i.e., first alphabet of each word).

Sample Input : LAL KRISHNA ADVANI

Sample output : L K A

7. Write a program in Java to accept a name containing three words and display the surname first followed by the first and middle names.

Sample Input : MOHANDAS KARAMCHAND GANDHI

Sample output : GANDHI MOHANDAS KARAMCHAND

8. Write a program in Java to accept a sentence and find the frequency of a given alphabet.

Sample Input : WE ARE LIVING IN COMPUTER WORLD.

: Enter an alphabet whose frequency is to be checked (Say: E).

Sample output : The frequency of 'E' is 3.

9. A man has written a statement as " My name is Alok Kumar Gupta and my age is 45 years". Later on he realized that he had declared his name as Alok instead of Ashok and the age 45 instead of 35. Write a program in Java to correct his age in the predicted statement. Display the output as : My name is Ashok Kumar Gupta and my age is 35 years.

10. Write a program in Java to enter a String/Sentence and display the longest word and the length of the longest word present in the String.

Sample Input : "TATA FOOTBALL ACADEMY WILL PLAY AGAINST MOHAN BAGAN"

Sample output : The longest word : FOOTBALL : The length of the word : 8

[ICSE 2009]

11. Write a program in Java to accept a String in upper case and find the frequency of each vowel present in the String.

Sample Input : "RAIN WATER HARVESTING ORGANISED BY JUSCO"

Sample output : Frequency of 'A' = 4
Frequency of 'E' = 3
Frequency of 'I' = 3
Frequency of 'O' = 2
Frequency of 'U' = 1

12. Write a program in Java to accept a String as :
"IF IT RAINS, YOU WILL NOT GO TO PLAY"
Convert all characters of the words in lower case other than the first characters so as to obtain the following output:

Sample output : If It Rains, You Will Not Go To Play

13. Write a program in Java to accept a word and display the ASCII of each character.

Sample Input : BLUEJ

Sample output : ASCII of B = 66
ASCII of L = 75
ASCII of U = 84
ASCII of E = 69
ASCII of J = 73

14. Write a program in Java to accept a String in upper case and replace all the vowels with Asterisk (*) present in the String.

Sample Input : "TATA STEEL IS IN JAMSHEDPUR"

Sample output : T*T* ST**L *S *N J*MSH*DP*R

15. Write a program in Java to enter a String and frame a word by joining all the first characters of each word. Display the new word.

Sample Input : Vital Information Resource Under Seize

Sample Output : VIRUS

16. Write a program in Java to enter a String and display all the palindrome words present in the String.

Sample Input : MOM AND DAD ARE NOT AT HOME

Sample Output : MOM

DAD

17. Write a program to accept a string. Display the string in a reversed order.

Sample Input : Computer is Fun

Sample Output : Fun is Computer

18. Write a program to input a string and print the character which occurs maximum number of times within a string.

Sample Input : James Gosling developed Java

Sample Output : The character with maximum number of times: e

19. Write a program to input a string and print the word containing maximum number of vowels.

Sample Input : HAPPY NEW YEAR

Sample Output : The word with maximum number of vowels: YEAR

20. Consider the string as:

Blue bottle is in Blue bag lying on Blue carpet

Write a program to accept a sample string and replace the word Blue with Red. The new string is displayed as:

Red bottle is in Red bag lying on Red carpet

21. A computer typist has the habit of deleting the middle name 'Kumar', while entering the names containing three words. Write a program to enter a name with three words and display the new name after deleting middle name 'Kumar'.

Sample Input : Ashish Kumar Nehra

Sample Output : Ashish Nehra

22. Write a program to accept a word and convert it into lower case, if it is in upper case. Display the new word by replacing only the vowels with the character following it.

Sample Input : computer

Sample Output : cpmpvtfr [ICSE 2011]

23. Write a program to accept a word and display the new word after removing all the repeated alphabets.

Sample input : applications

Sample Output : aplictons

24. A string is said to be 'Unique', if none of the alphabets present in the string are repeated. Write a program to accept a string and check whether the string is Unique or not. The program displays a message accordingly.

Sample Input : COMPUTER

Sample Output : Unique String

25. Write a program to input a string and display the frequency of vowels of each word present in the string.

Sample Input : MASTERING INFORMATION TECHNOLOGY

Sample Output : Number of vowels present in MASTERING = 3

Number of vowels present in INFORMATION = 5

Number of vowels present in TECHNOLOGY= 3

26. A non-palindrome word can be made a palindrome word just by adding reverse of the word with the original word. Write a program to accept a non-palindrome word and display the new word after making it a palindrome.

Sample Input : ICSE

Sample Output : The new word after making it palindrome as:

ICSEESCI

27. Write a program to input a String. Count and display the frequency of each alphabet in an order, which is present in the String. [ICSE 2010]

Sample Input : COMPUTER APPLICATIONS

Sample Output :

Character	Frequency	Character	Frequency
A	2	O	2
C	2	P	3
I	1	R	1
L	2	S	1
M	1	T	2
N	1	U	1

28. Write a program to accept a string. Convert the string to upper case. Count and output the number of double letter sequences that exist in the string.

Sample Input : "SHE WAS FEEDING THE LITTLE RABBIT WITH AN APPLE"

Sample Output : 4 [ICSE 2012]

29. Write a program to accept a word (say, BLUEJ) and display the pattern:

(a)
```
B L U E J
B L U E
B L U
B L
B
```

(b)
```
J
E E
U U U
L L L L
B B B B B
```

(c)
```
B L U E J
L U E J
U E J
E J
J
```

30. Write a program to display the pattern:

(a)
```
A B C D E
B C D E
C D E
D E
F
```

(b)
```
A
B C
D E F
G H I J
K L M N O
```

(c)
```
A B C D E
A B C D A
A B C A B
A B A B C
A A B C D
```

(d)
```
A B C D E
  B C D E
    C D E
      D E
        E
```

(e)
```
E D C B A
  E D C B
    E D C
      E D
        E
```

(f)
```
        A
      A B
    A B C
  A B C D
A B C D E
```

31. Write a program to generate a triangle or an inverted triangle till n terms based upon User's choice of triangle to be displayed. [ICSE MODEL]

Example 1:

Input: Type 1 for a triangle and

Type 2 for an inverted triangle

Enter your choice 1

Enter the number of terms 5

Sample Output:

```
*  *  *  *  *
   *  *  *  *
      *  *  *
         *  *
            *
```

Example 2:

Input: Type 1 for a triangle and

Type 2 for an inverted triangle of alphabets

Enter your choice 2

Enter the number of terms 5

```
A B C D E
A B C D
A B C
A B
A
```

32. Write a program to generate a triangle or an inverted triangle based upon User's choice. [ICSE MODEL]

Example 1:

Input: Type 1 for a triangle and

Type 2 for an inverted triangle

Enter your choice 1

Enter a word : BLUEJ

Sample Output:

```
        B
      L L
    U U U
  E E E E
J J J J J
```

Example 2:

Input: Type 1 for a triangle and

Type 2 for an inverted triangle

Enter your choice 2

Enter a word : BLUEJ

```
B L U E J
B L U E
B L U
B L
B
```